

Vector Store Serialization and Cache Acceleration

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Module: VectorStoreCache

File Path: backend/app/services/vector_stores/vector_store_cache.py

1. Cache Mechanics

To eliminate redundant document parsing and costly embedding API calls, MCP PRO implements deterministic file-backed vector caching.

- Cache Key: SHA-256 hash of document_url combined with loader variant (e.g. doc_url::std vs doc_url::llm).
- Storage Format: LangChain vector store serialized dump files stored in vector_store_cache/*.vs.
- Ledger: cache_metadata.json tracks URL mappings, file paths, and creation timestamps.

2. Benchmark Metrics

- Cold Request: Involves network download, text extraction, recursive splitting, embedding generation, and vector DB insertion. Typical latency: 1.5s - 8.0s depending on document size.
- Warm Request: Direct deserialization from local disk. Typical latency: 5ms - 25ms.
- Latency Reduction: Consistently achieves > 90% reduction in indexing time.